

Section 20 Solution

Anson Tam and Bruno Smaniotto*

2 Exercises

2.1 Moral Hazard

Consider the moral hazard (hidden action) problem which we considered in lecture. An agent is offered a contract $w = a + by$, where w is the wage, and y is the output, with $y = e + \varepsilon$, where e is (unobservable) effort and ε is noise with mean 0 and variance σ^2 . Remember that the risk-averse agent (with constant risk aversion parameter γ) has expected utility of a contract being equal to¹

$$EU(e) = E[u(w) - c(e)] = E(w) - \frac{\gamma}{2} \text{var}(w) - c(e) = a + be - \frac{\gamma}{2} b^2 \sigma^2 - c(e)$$

And the risk-neutral firm has expected profits

$$E\pi = E(y) - E(w) = e - E(w) = e - a - be$$

1. Interpret the terms in $EU(e)$.

Answer:

- $a + be$ gives the expected wage from the contract. a is the base pay independent of e , but b is the incentive pay: it increases in e in expectation, creating an incentive for workers to work more.
 - Because the agent is risk averse, variance of wage $b^2 \sigma^2$ is deducted due to the uncertainty in output. The uncertainty in wage is determined by both the level of noise (ε) and the degree to which effort is related to noise as specified by the contract (b). In addition, agent's exposure to the noise is scaled by a factor in proportion to the coefficient of absolute risk aversion, $\frac{\gamma}{2}$.
 - $c(e)$ is the effort cost.
2. Remember that effort is costly with cost of effort $c(e)$, which satisfies $c'(e) > 0$ and $c''(e) > 0$ for all e . Without solving for the overall problem, explain intuitively, but also as clearly as you can, why the firm will not set $b^* = 0$ in the optimum. What effort would workers choose for $b = 0$?

Answer:

If $b = 0$, the agent does not increase his expected utility with more effort, and will choose $e = 0$, because effort is costly. By exposing the agent to changes in output ($b > 0$), the principle incentivizes effort (but also creates uncertainty on the wage!).

3. In light of this, explain qualitatively what it means that the firm is facing a risk-incentive trade-off in setting the optimal piece rate b^* .

Answer:

Holding a^* constant. In terms of risks, a higher b implies that the firm passes a high level of uncertainty to the agent and reduces the incentive for the agent to take up the contract, which the firm does not like. In terms of incentives, a higher b incentivizes the worker to have higher e , which leads to higher y and π , which the firm prefers. As a result, there is a risk-incentive tradeoff in setting b^* .

*These notes are a consolidation of those of Jianmeng Lyu, Sam Leone, Pablo Munoz, Yassine Sbair Sassi, Jon Schellenberg, Nick Li, Katalin Springel, Jonas Tungodden, Dan Acland, Justin Gallagher, Mariana Carrera, Matt Leister, Ivan Balbuzanov, Anne Karing, and David Wu, Sam Wang, and Matteo Saccarola.

¹ $E[u(w)] = E(w) - \frac{\gamma}{2} \text{var}(w)$ is a result of the exponential utility $u(w)$. If you are interested in the derivation, see for example: <https://www.tau.ac.il/~spiegel/teaching/corpfin/mean-variance.pdf>

4. Now, let's quickly solve an easy version of this model, using our friend Backward Induction:

- (a) Suppose $c(e) = \bar{c}e^2/2$ for some constant $\bar{c}$. Taking the wage contract as given, what is the level of effort that maximizes the agent's expected utility?

Answer:

$$\max_e EU(e) = \max_e a + be - \frac{\gamma}{2}b^2\sigma^2 - \bar{c}e^2/2$$

FOC gives

$$b - \bar{c}e^* = 0 \Rightarrow \boxed{e^* = b/\bar{c}}$$

- (b) Suppose the agent has a reservation utility of $\bar{u}$. (This is different from lecture! In lecture, we just wanted utility to be non-negative). What is the agent's participation (aka individual rationality) condition, given e^* and wage $w(y)$? That is, what condition must be satisfied for the worker to take the contract?

Answer: For our agent to take the contract, we need the following to hold:

$$\begin{aligned} EU(e^*) &\geq \bar{u} \\ a + be^* - \frac{\gamma}{2}b^2\sigma^2 - \bar{c}(e^*)^2/2 &\geq \bar{u} \\ a &\geq \bar{u} - be^* + \frac{\gamma}{2}b^2\sigma^2 + \bar{c}(e^*)^2/2 \\ \Rightarrow a &= \bar{u} - be^* + \frac{\gamma}{2}b^2\sigma^2 + \bar{c}(e^*)^2/2 \end{aligned}$$

The inequality will hold with equality because the firm will want a as low as possible (as seen from their objective function).

- (c) Recall that the shareholder wants to maximize expected profits $\mathbb{E}[\pi] = e - a - be$. Solve for the optimal a and b given e^* , the individual rationality condition, and the other parameters in this problem (namely γ , σ^2 , $\bar{c}$, and $\bar{u}$).

Answer:

Formally, the firm's maximization problem is as follows:

$$\max_{a,b} e^* - a - be^* \text{ s.t. } e^* = b/\bar{c} \text{ and } a = \bar{u} - be^* + \frac{\gamma}{2}b^2\sigma^2 + \bar{c}(e^*)^2/2$$

Substituting these constraints in (using the fact that the individual rationality constraint will hold with equality), we can rewrite this problem as follows:

$$\max_b (b/\bar{c}) - \left[\bar{u} - b(b/\bar{c}) + \frac{\gamma}{2}b^2\sigma^2 + \bar{c}(b/\bar{c})^2/2 \right] - b(b/\bar{c})$$

We can differentiate with respect to b and set equal to 0, so we get

$$\frac{1}{\bar{c}} + 2\frac{b^*}{\bar{c}} - \gamma b^*\sigma^2 - \frac{b^*}{\bar{c}} - 2\frac{b^*}{\bar{c}} = 0 \Rightarrow \boxed{b^* = \frac{1}{1+\gamma\sigma^2\bar{c}}}$$

We can then plug b^* into e^* (optimal effort condition) to solve for e^* ; plug b^* into e^* into the participation (individual rationality) condition to solve a^* . And plug them back into objective functions to obtain expected profits and expected utility.

2.2 Adverse Selection (Problem 2, Final, Spring 2009)

Wisdom teeth are usually removed early in life as a preventive measure, to avoid problems that may happen in the future. It is more costly to remove the teeth once the problems have begun.

Suppose there are two types of wisdom teeth: those that are more likely to lead to problems in the future ("bad" teeth) and those that are less likely to lead to problems ("good" teeth). Specifically, assume a good set of teeth has a 10% chance of problems and a bad set of teeth has a 60% chance of problems. All future problems will cost \$500 to solve. However, if the wisdom teeth are removed early in life the probability of

future problems is 0. There are N patients with wisdom teeth in Berkeley.

Assume the consumer does not discount between periods and has utility directly equal to his wealth: $u(\$x) = x$. Notice that this means he is risk-neutral.

Suppose there is only one oral surgeon in Berkeley, and she operates as a monopolist in wisdom tooth removal. She faces marginal cost \$100 of removing any set of wisdom teeth, whether they are good or bad teeth.

1. How much is an individual willing to pay for the removal of a good set of wisdom teeth? And for the removal of a bad set of wisdom teeth?

Answer:

Good tooth: Expected cost without removing the tooth is $.10 \cdot 500 + .90 \cdot 0 = 50$, so she is willing to pay at most \$50 to remove a good tooth and have an expected cost of 0. For future reference, let's denote this value $V_g = 50$

Bad tooth: Expected cost without removing the tooth is $.60 \cdot 500 + .40 \cdot 0 = 300$, so she is willing to pay at most \$300 to remove a bad tooth and have an expected cost of 0. Let's denote this value $V_b = 300$

2. First-best scenario: If consumers can identify whether their wisdom teeth are good or bad, what will the oral surgeon monopolist set as the price for removing wisdom teeth (p_{FB}), and which type(s) of teeth will be removed?

Answer:

The monopolist will not remove a *set of* wisdom teeth if she gets paid less than her marginal cost for doing so, so no good teeth will be removed. Since she is only removing bad teeth, she can charge the consumers' full willingness to pay for bad tooth removal: \$300 per tooth.

3. How much profit will the oral surgeon monopolist make per set of wisdom teeth?

Answer:

Profit will be \$200 per *set of* wisdom teeth, but only bad teeth will be removed.

4. Hidden type scenario: Now, suppose individuals cannot tell whether their teeth are good or bad, but they know that across the population, $\frac{2}{5}$ of all peoples teeth are "bad". Solve for the expected benefit to an individual of removing his wisdom teeth, given his (correct) beliefs about the likelihood of each type.

Answer:

Expected benefit of removing a tooth of unknown quality is $\pi_g \cdot V_g + \pi_b \cdot V_b$, where π_g is the probability the tooth is good, and π_b is the probability the tooth is bad. With the value given, $\pi_b = 2/5$, this works out to be: $\frac{3}{5}50 + \frac{2}{5}300 = 150$.

5. Given the expected benefit you found above, what price (p_{HT}) will the monopolist charge to remove a set of wisdom teeth when teeth have hidden type? Which type(s) of teeth will be removed in equilibrium?

Answer:

Since this value is greater than the marginal cost of removing a tooth, the monopolist will charge \$150 (again extracting all the surplus) and all wisdom teeth will be removed.

6. Calculate the profit of the monopolist in this case (hidden type) and compare it to his profits in the case above (first best). (Remember that there are N patients, and $\frac{2}{5}$ of them have bad wisdom teeth.) Which case does she prefer? Interpret.

Answer:

Hidden type: Profit is \$50 per tooth, but all teeth are removed. If total number of people is N , monopolist makes profit $50N$.

First-best: Profit is \$200 per tooth, but only the bad teeth are removed, so total profit is $200 \cdot \frac{2}{5}N = 80N$. The monopolist prefers the the first-best case, because even though she gets less business, the price of removing teeth is higher and makes greater total profit.

7. What about the consumers? compare the aggregate expected utility of all N consumers in the hidden type case with the aggregate in the first-best case. Which case is better, in aggregate?

Answer:

Hidden type: $EU = w_0 - 150$.

First best: $EU = \frac{2}{5} \cdot (w_0 - 300) + \frac{3}{5} \cdot (\frac{1}{10}(w_0 - 500) + \frac{9}{10}w_0) = w_0 - 150$

It should not surprise us that the consumer is equally well off in both cases, since the monopolist has been able to charge him exactly his maximum willingness to pay in both cases. Moreover, if you calculate the EU in the scenario without dentist you will get the same answer! The dentist, as a monopolist, is able to extract all the surplus from the transaction.